

Enhancing radio foreground models with observations from lunar orbit

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ABSTRACT

Radio foreground models are important ingredients to a wide range of astronomical analyses, but are known to have significant flaws, ranging from unrealistic spectral behaviour to large flux scale calibration errors of tens of percent. Some of the issues can be traced directly to the input data used to calibrate the models, such as all-sky maps with uncertain calibration and noise characteristics. While it may be possible to make new precision-calibrated maps from the ground using new telescopes such as 21cm global signal experiments, these are limited in resolution, sky coverage, and frequency coverage due to terrestrial radio frequency interference. In this paper, we simulate the possibility of recalibrating existing radio sky models using a radio telescope in lunar orbit to make precision-calibrated full-sky observations that are free from terrestrial RFI in the 1.5 – 2 GHz window. Using a high-dimensional Bayesian sky model inference method, we show that...

Key words: data methods – cosmology: observations

1 INTRODUCTION

Earth’s atmosphere provides two sizeable spectral windows, at radio and optical wavelengths, that have enabled the development of terrestrial astronomical facilities. In other frequency bands, atmospheric absorption or reflection necessitates airborne or spaceborne observing, which only became feasible in the latter half of the 20th Century (Giacconi et al. 1968; Walker 2000; Linsky 2018; Santangelo et al. 2023). Terrestrial facilities have significant advantages in terms of cost, serviceability, and accessibility. In the 21st Century, the traditional terrestrial windows have come under threat from human-made factors however. Satellite constellations, now growing to fleets with tens of thousands of individual spacecraft, significantly degrade ground-based optical imaging through the sheer number of satellite trails crossing long-exposure images (Walker et al. 2020; Bassa et al. 2022; Barentine et al. 2023). They are also sources of radio-frequency interference (RFI) in their designated transmit/receive bands, electromagnetic interference (EMI) in other bands from on-board electronics, and reflected terrestrial radio emission from over the horizon (Zhang et al. 2018; Di Vruno et al. 2023; Peel et al. 2025; De Pree et al. 2026). The number of such systems continues to increase.

At the same time, ground-based spectrum usage (e.g. for telecommunications) continues to expand. At radio wavelengths, large chunks of the sub-GHz band are taken up by digital TV and radio broadcasts, while cellular phones, wireless networking, and GNSS systems heavily pollute the GHz band and above (Harper & Dickinson 2018; Saide et al. 2023; Engelbrecht et al. 2025). Orbital extensions to these networks are also now coming online, making it hard to evade severe RFI even by placing astronomical facilities in remote, sparsely-populated areas (Agrawal et al. 2025). Bright, time-variable artificial emissions can cause significant problems for radio astronomy facilities even when they are far away, heavily filtered, or outside the design band of

the instrument (Offringa et al. 2013). As low-noise, ultra-wideband receivers are becoming widely available, their usefulness is being constrained by an ever-worsening RFI, despite efforts by the ITU and other international organisations to protect important parts of the spectrum (DeBoer et al. 2013).

An alarming prospect is that we may be witnessing the gradual closure of Earth’s terrestrial observing windows due to the inexorable growth of artificial contamination. Technical mitigations, such as increasingly sophisticated flagging, masking, and filtering methods (e.g. Offringa et al. 2012; Wilensky et al. 2020; Morello et al. 2022), may eventually reach a point of diminishing returns. In some cases, even shifting to space-based observing may be insufficient, as RFI remains extremely bright in Earth orbit and beyond (Saide et al. 2023). A particular risk is that certain types of observation, and therefore certain astronomical phenomena, may be put beyond our reach in any practical sense, effectively closing several windows on the Universe.

An important example is the redshifted 21cm emission from Cosmic Dawn and the Dark Ages. The sky-averaged temperature and brightness temperature anisotropies encode important and unique information about the formation of the first stars and galaxies, possible signatures of exotic particles such as certain dark matter candidates, and information about inflationary processes that is otherwise impossible to recover through other means (Pritchard & Loeb 2012; Liu et al. 2022; Mondal & Barkana 2023). At the lowest frequencies (highest redshifts), below tens of MHz, the ionosphere already strongly attenuates astronomical signals (Datta et al. 2014), and so space-based observing is a necessity for high-redshift Dark Ages studies. Even with ionospheric attenuation, terrestrial RFI remains a serious contaminant for space-based experiments, and so various proposals have been made to develop Dark Ages observatories that operate on the far-side of the Moon, i.e. one of the few remaining places where strong attenuation of terrestrial RFI can be guaranteed (Bassett et al. 2020; Burns et al. 2021; Smith & Pober 2025).

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In this paper, we consider another observing window that is already heavily contaminated from the ground and in Earth orbit – the decimetre, or UHF, band ($\sim 300 - 3000$ GHz). Observations in this band have a rich history in radio astronomy, particularly due to the presence of the rest-frame 21cm line at 1.42 GHz, but also due to the steep spectrum of radio synchrotron emission from the Galaxy and extragalactic sources that increases towards lower frequencies. Full-sky maps in this window provide templates for modelling contaminating foreground emission at both lower and higher frequencies where, respectively, 21cm surveys and CMB experiments operate (de Oliveira-Costa et al. 2008). One of the main examples is the full-sky map at 408 MHz of Haslam et al. (1982), which was observed with sub-degree angular resolution using three different telescopes over more than a decade. There are known to be uncorrected residual systematic effects in the 408 MHz map however. Issues such as striping artifacts and improperly subtracted point sources have been addressed in an improved map by Remazeilles et al. (2015), but substantial spatially-varying calibration errors still remain, potentially at the tens of percent level (Wilensky et al. 2025). Additional observations with an improved absolute flux scale calibration are needed to further correct for these issues (e.g. Ignatov & Pritchard 2026; Carter et al. 2025; Nasirudin & Bull 2025a). Depending on the nature of the spatial variations in the calibration, these would also need to cover a large portion of the sky.

The raw data for the 408 MHz survey are no longer available, and this observing frequency is no longer protected, with bright RFI in immediately adjacent bands. Directly replicating the Haslam et al. (1982) survey is therefore no longer feasible from the ground. Observing in cleaner bands at lower frequency, e.g. around 300 MHz, requires very large (~ 100 m diameter) telescopes to approach 1° angular resolution however. Several recent studies (e.g. Carter et al. 2025; Nasirudin & Bull 2025b; Ignatov & Pritchard 2026) have considered the possibility of using 21cm global experiments, with absolutely-calibrated temperature scales but much worse (tens of degrees) angular resolution, to correct for some of the systematic effects observed in the 408 MHz map. These are limited in how well they can model the spatial variations however.

An ideal follow-up survey would potentially look quite different from the original Haslam et al. (1982) survey. An important innovation would be to use a large bandwidth (or multiple well-spaced observing bands) in order to constrain the spatially-varying spectral index of the Galactic and extragalactic emission as well. Moving to higher frequencies, between $\sim 500 - 1000$ MHz is arguably more plausible, but depending on the observing site there are wide bands contaminated by terrestrial RFI such as digital TV and cellular phones. For longer-term surveys, the prospect of the RFI environment degrading must also be considered, as new parts of the spectrum begin to be used for other purposes.

At the lower end of the band, cellular phone systems (e.g. GSM) and GNSS systems are persistently bright, and transient emission and reflection from aircraft is observed (Sihlangu et al. 2022). At higher frequencies, 2.4 GHz wireless networks have become ubiquitous. Substantial usable chunks of spectrum remain (e.g. MeerKLASS), but their bandwidth is limited by the intervening bright RFI.

This paper is organised as follows. In Sect. 2, we outline a space mission concept to make precision-calibrated full-sky maps of radio emission from $\sim 1.5 - 2$ GHz from the far-side of the Moon. In Sect. 3, we review the Bayesian sky model inference framework from Nasirudin & Bull (2025a), explaining how it can be used to recalibrate existing maps with spatially-varying calibration errors using data from the proposed space mission. We present simulated analysis results in Sect. 4, before concluding in Sect. 5.

2 OBSERVATIONAL PARAMETERS

In this section, we propose a basic space mission concept to make full-sky maps of radio emission at $\gtrsim 1$ GHz with good spectral resolution, high calibration precision, and avoiding worsening terrestrial RFI by observing from the far-side of the Moon. This is intended to fill an important gap between high-resolution microwave observations by CMB missions above 20 GHz; low-resolution, partial-sky (but absolutely-calibrated) 21cm global signal experiments at ~ 100 MHz; and the roughly degree-resolution Haslam et al. (1982) and Reich & Reich (1988) maps at 408 MHz and 1420 MHz respectively. The data from such a mission would have a high legacy value, particularly for improving radio and microwave sky models that are widely used in cosmology and other contexts.

2.1 Frequency band comparison

Simple calculations to compare different frequency ranges in nailing down spectral parameters. Also look at RFI expected in these bands.

Operating below about 1 GHz, a space-based instrument would require a very large antenna to achieve even moderate angular resolution, while at frequencies of several GHz, it is arguably more effective to build a terrestrial experiment.

To identify suitable frequency bands, we consider a simple per-pixel spectral model based on a moment expansion around a mean power-law spectral index β_0 , which takes the form

$$T_p(\nu) = g(\nu)T_{p,0} \left(\frac{\nu}{\nu_0} \right)^{\beta_{p,0}} \left(1 + \sum_{n=1}^{N_m} m_n^{(p)} [\ln(\nu/\nu_0)]^n \right) + T_{\text{offset}}(\nu). \quad (1)$$

Here, $T_{p,0}$ is the brightness temperature for pixel p at the pivot frequency ν_0 , and $m_n^{(p)}$ are coefficients of each term in the moment expansion at this pixel. The mean spectral index can either vary with pixel or be chosen to remain constant across the sky, in which case $\beta_{p,0} = \beta_0$. Calibration errors are incorporated by a multiplicative scale factor, $g(\nu)$, and a zero-point offset at each frequency, $T_{\text{offset}}(\nu)$.

We next consider a set of measurements of the same pixel at different frequencies. We ignore the impact of different beam sizes at this stage. The measurement uncertainty on each data point is assumed Gaussian, and so we can write the likelihood as

$$\ln \mathcal{L}(\vec{d}|\vec{\theta}) = -\frac{1}{2} [\vec{d} - \vec{m}(\vec{\theta})]^T N^{-1} [\vec{d} - \vec{m}(\vec{\theta})] + c, \quad (2)$$

where $\vec{d}$ is a vector of measurements of T_p at different frequencies, $\vec{m}$ is a vector of model predictions from Eq. 1 given a set of parameter values $\vec{\theta}$, N is the frequency-frequency covariance matrix of the measurements, and c is a normalisation factor.

The Fisher matrix formalism can be used to make simplified forecasts for how well each parameter can be measured given a particular experimental setup. For the Gaussian likelihood in Eq. 2, the Fisher matrix simplifies to

$$F_{ab} = \frac{\partial \vec{m}^T}{\partial \theta_a} N^{-1} \frac{\partial \vec{m}}{\partial \theta_b}, \quad (3)$$

where a and b label the parameters of the model, and the partial derivatives are evaluated at a particular set of ‘fiducial’ model parameters, θ_{fid} . Calculating this matrix and inverting gives an estimate of the covariance matrix of the parameters that would be expected from fitting the model to a typical data vector, i.e. $C_{ab} \approx (F^{-1})_{ab}$.

We now use this formalism to compare different configurations of frequency bands. For the fiducial model, we assume that the calibration parameters are fixed to $g = 1$ and $T_{\text{offset}} = 0$ K except

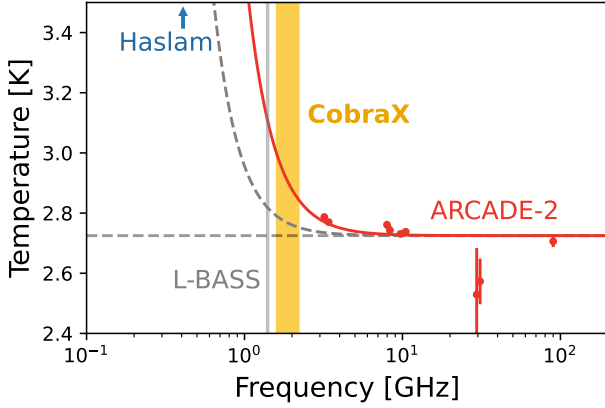


Figure 1. xxx.

from a single data point at 408 MHz, for which $g_{408} = 1.2$ and $T_{\text{offset},408} = 1$ K, modelling the possible systematic offsets in the Haslam map. We consider a single pixel with $T_{p,0} = 2$ K at a pivot frequency of 1 GHz, $\beta_{p,0} = -2.75$, and two moments with fiducial values $m_1^{(p)} = m_2^{(p)} = 0$. This results in six free parameters per pixel: $\{g_{408}, T_{\text{offset},408}, T_{p,0}, \beta_{p,0}, m_1^{(p)}, m_2^{(p)}\}$.

The observing setup assumes that suitably beam-convolved Haslam 408 MHz data are included in the model fits, along with N_{bands} new coarse frequency channels from the satellite instrument. These are taken to have constant width $\delta\nu$ and minimum and maximum frequencies ν_{min} and ν_{max} . We assume a constant noise rms per frequency channel given by the radiometer equation as

$$\sigma_\nu = \frac{T_{\text{sys}}}{\sqrt{\delta\nu t_{\text{obs}}}}, \quad (4)$$

with $T_{\text{sys}} = 150$ K, and a total observing time of 2500 h (corresponding to a duty cycle of about 30% in 1 year). For 6° pixels and a perfectly uniform observing strategy, this equates to an observing time per pixel of $t_{\text{obs}} = 2.2$ h.

Fig. XXX shows the predicted standard deviation on each parameter for several scenarios. We consider four centre frequencies: 600 MHz, 1000 MHz, 1400 MHz, and 1800 MHz. For each of these, we consider either a ‘narrow’ bandwidth of 200 MHz (‘N’), or a ‘wide’ bandwidth of 400 MHz (‘W’). In each of these cases, we consider either ‘fine’ frequency channels of 10 MHz (‘F’), or ‘coarse’ frequency channels of 25 MHz (‘C’). Each case is labelled according to these letters, so for example ‘1400WF’ is centred at 1400 MHz with a wide 400 MHz bandwidth and fine channels of 10 MHz.

2.2 Angular and spectral resolution

Simple calculations to determine useful ranges of angular scales and spectral scales (we can do simple Fisher forecasts for this).

At intermediate frequencies, the angular resolution of the instrument is limited by several constraints. At 2 GHz, an antenna aperture of ~ 8.5 m would be required to achieve $\sim 1^\circ$ resolution, increasing to 17 m at 1 GHz. These dimensions are potentially achievable with deployable structures, but at great cost, while the survey speed (measured as the sky area surveyed per unit time) would also be reduced. A middling resolution of around 10° would permit small-scale (CubeSat-compatible) deployable antenna options and a shorter mission duration, at the expense of being insensitive to small-scale calibration issues in the low-frequency foreground maps. We consider

three cases in what follows: beams with full-width at half-maximum (FWHM) of 2° , 6° , and 12° at 1.5 GHz, respectively.

Fig. XXX shows an example of the RFI environment in the 1 – 2 GHz band, as observed from Jodrell Bank Observatory. The contamination is still quite bad even in ostensibly radio-quiet locations such as the MeerKAT site in South Africa.

2.3 Lunar orbital considerations

The major advantage of placing a telescope in lunar orbit is the strong suppression of terrestrial radio-frequency interference (RFI) expected on the lunar far-side, so long as the telescope is in the shadow of the Moon. At very low frequencies of a few MHz and below, diffraction around the lunar limb reduces the size of the shadow region, and effects such as the irregular surface geometry and dielectric properties of the lunar regolith must be taken into account (Bassett et al. 2020). At GHz frequencies diffraction effects are negligible however, and the shadow region is geometrically defined, as shown in Fig. 2. We will assume that strong RFI suppression is achieved whenever a spacecraft is within this region. Effects due to relay satellites and emitters either in lunar orbit or on the lunar surface are ignored, although these are likely to become an important source of RFI even into the shadow region as lunar activities expand in the coming decade.

Referring to Fig. 2, we define d_{E-M} as the distance between the Earth and Moon centroids at a given time; R_E and R_M to be the (spherical-average) Earth and Moon radii respectively; and h_{geo} to be the average altitude of geostationary orbits, we can solve for the shadow angle θ via

$$\tan \theta = (R_E + h_{\text{geo}} - R_M)/d_{E-M}, \quad (5)$$

and hence derive an expression for the distance of the shadow region boundary from the Moon centroid,

$$h_{\text{shadow}}(\alpha) = R_M \frac{\cos \theta}{\sin(\alpha + \theta)}, \quad (6)$$

as a function of the angle off the Earth-Moon axis with respect to the Moon centroid, α . The Earth-Moon distance d_{E-M} is a function of time, and varies between 363,104 – 405,696 km. The geostationary orbit altitude is taken as 35,786 km, while the Earth and Moon radii are taken to be 6,371 km and 1,737 km respectively.

Fig. 3 shows the amount of time a spacecraft would spend in the shadow region if it is placed on a simple Keplerian orbit around the Moon in the plane of Fig. 2. The orbital parameters are varied using the initial altitude above the lunar surface and the initial transverse velocity at the anti-Earth point (the point of the orbit along the Earth-Moon centroid vector that is on the far-side of the Moon). These set the ellipticity and semi-major axis of the orbit, which is assumed stable. In fact, due to highly irregular mass concentrations (‘mascons’) within the Moon, low-altitude lunar orbits (below about 100 km) are unstable, with significant perturbations to the simple Keplerian picture. Orbits above this are also perturbed by the mascons. So-called ‘frozen orbits’ exist, which are stable for certain orbital inclinations, even at low altitude. We do not take these complications into account here however.

In this simple orbital picture, the time spent in the shadow region is maximised for the lowest altitude orbits, where the point of closest approach to the Moon (the ‘aposelene’) is on the Earth side.

3 BAYESIAN FOREGROUND INFERENCE FRAMEWORK

Recap the Gibbs sampler and discuss any changes/developments.

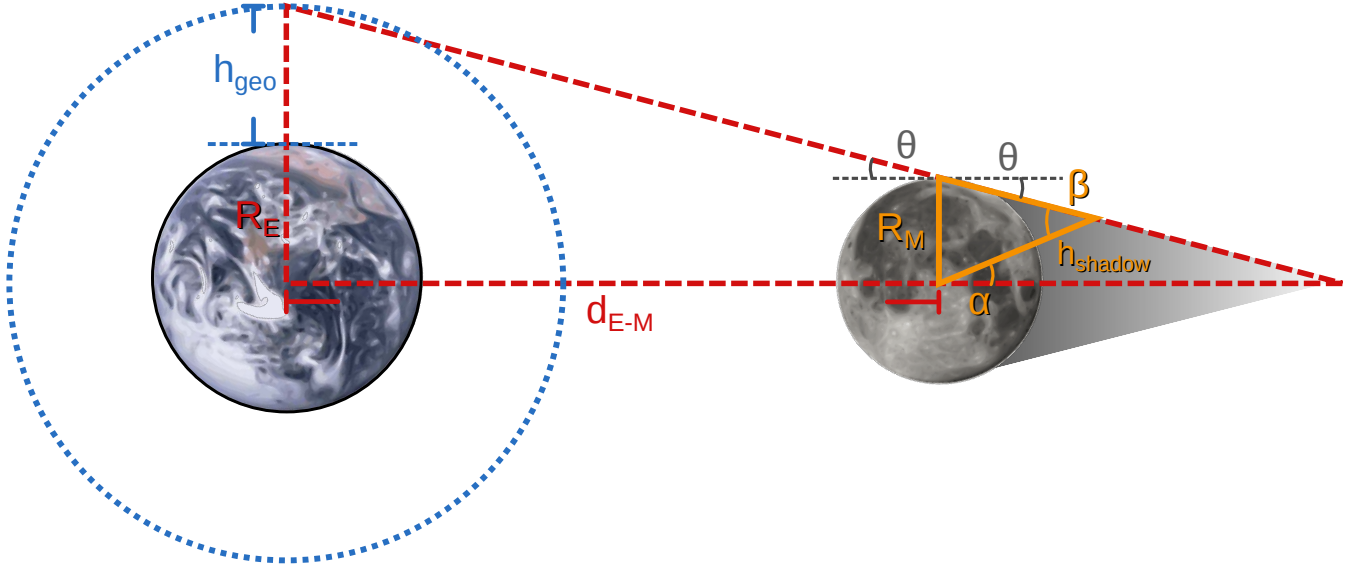


Figure 2. Simplified geometry of the Earth-Moon system, used to define an RFI shadow region on the lunar far-side. Geosynchronous orbit is used to define the extent of the RFI-emission envelope around Earth.

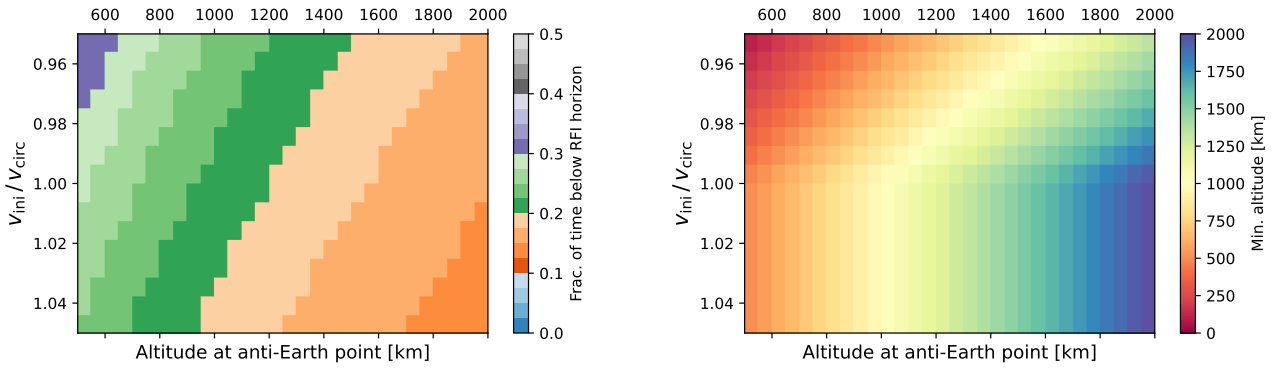


Figure 3. (Left): The fraction of time a spacecraft in a simplified Keplerian orbit around the Moon will spend in the RFI shadow region (i.e. below the ‘RFI horizon’), as a function of its initial tangential velocity and altitude at the point of its orbit farthest from Earth. (Right): The corresponding minimum altitude of the orbits above the lunar surface.

4 RESULTS

4.1 Single-band vs multi-band recalibration of the Haslam 408 MHz map

Plot of recovered gain error vs noise level and vs angular resolution, for single frequency band. Then compare to same cases with N bands.

4.2 Incorporation of 21cm global signal experiment data

Repeat the analyses above, but now with idealised 21cm global signal data added.

4.3 Excess radio background

Simple forecast for recovery of an excess radio synchrotron background monopole using these data.

5 CONCLUSIONS

ACKNOWLEDGEMENTS

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DATA AVAILABILITY

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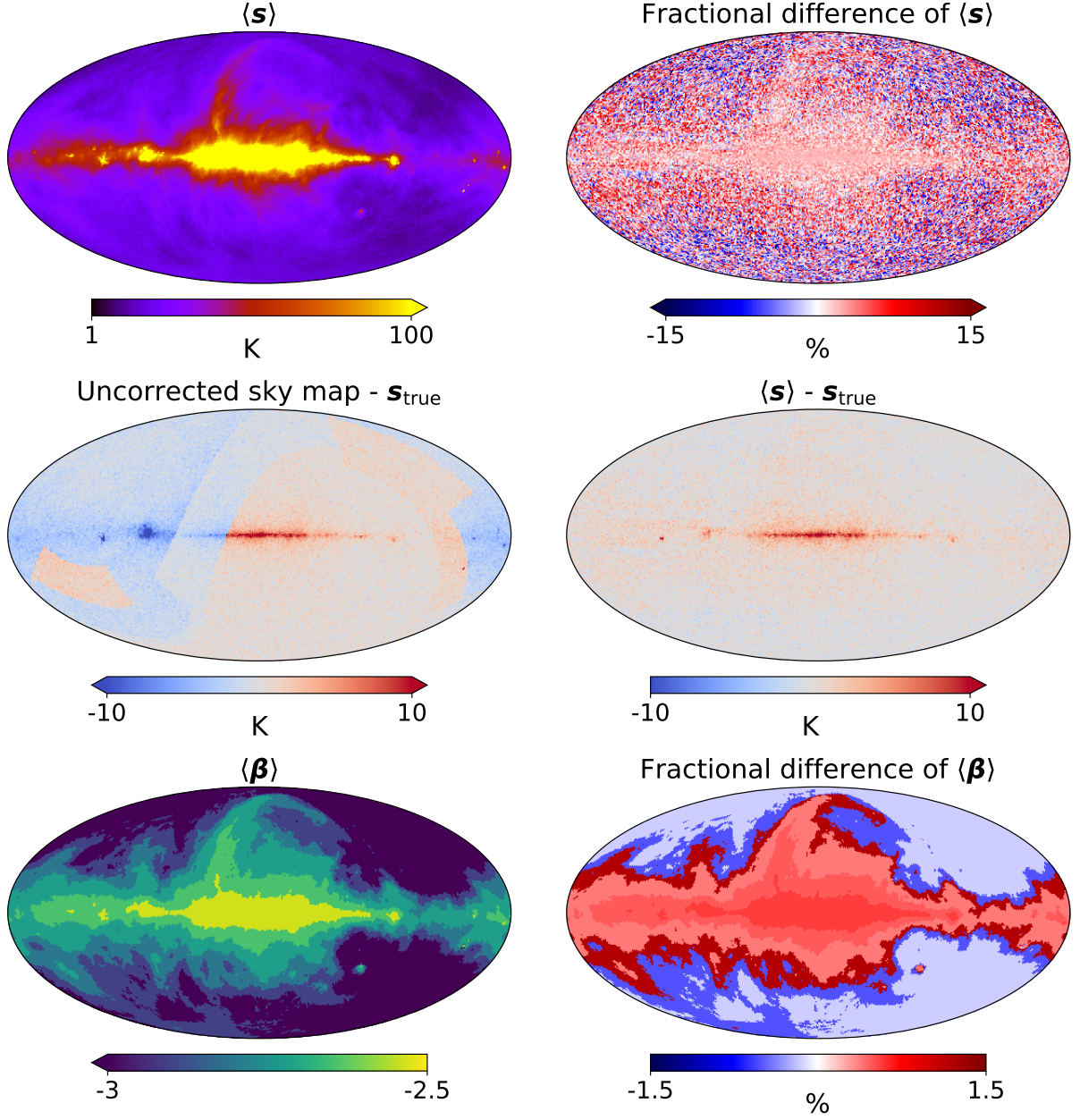


Figure 4. The posterior mean of the sky brightness temperature and its fractional difference with respect to the true map (top right), as well as the difference between the uncorrected sky map (middle left) and posterior mean (middle right) with the true sky brightness temperature field, and the posterior mean of the spectral index $\langle \beta \rangle$ (bottom left) and its fractional difference with respect to the true values (bottom right) with multi-band data between 1.55 to 2.05 GHz.

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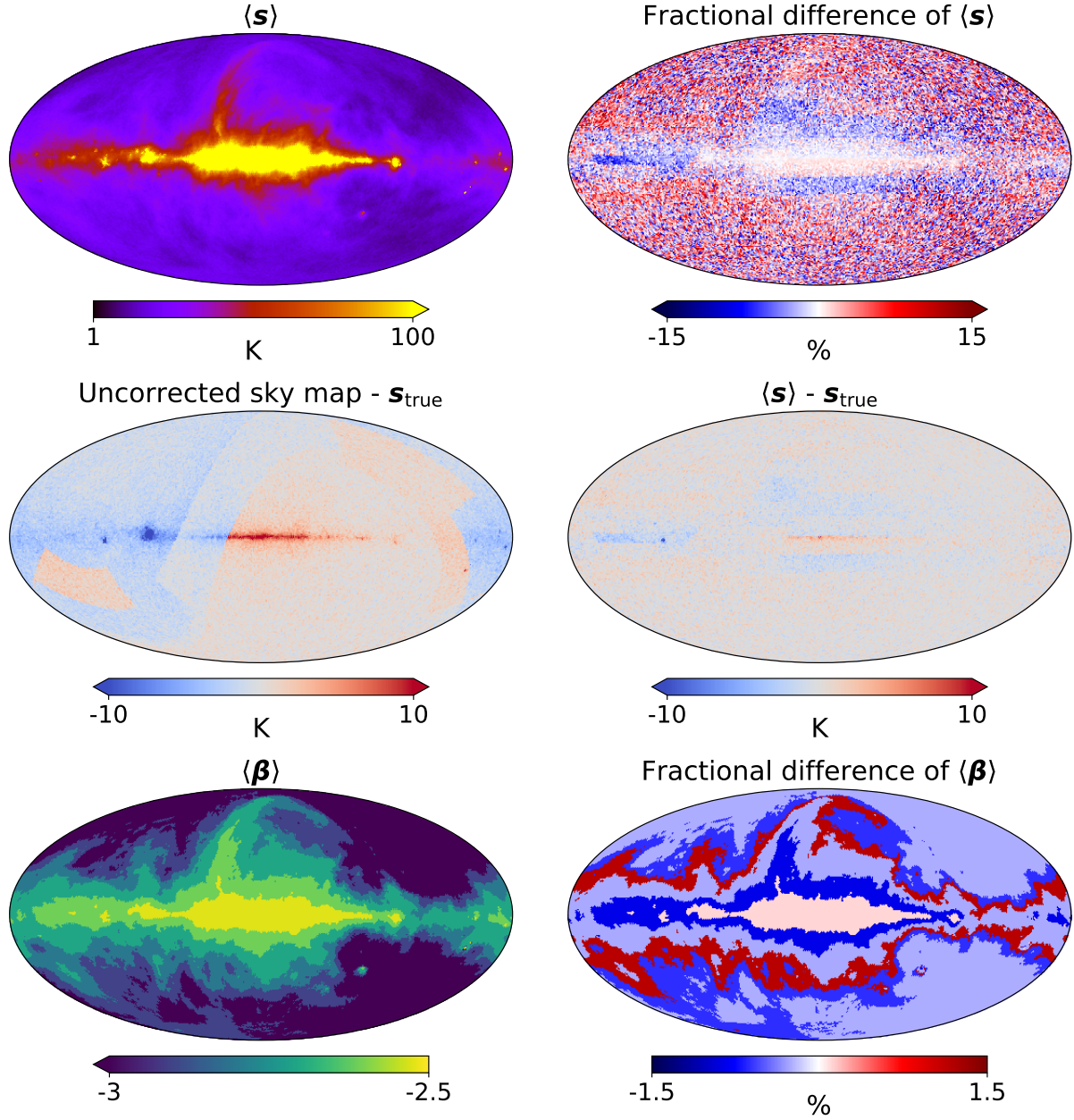


Figure 5. The posterior mean of the sky brightness temperature and its fractional difference with respect to the true map (top right), as well as the difference between the uncorrected sky map (middle left) and posterior mean (middle right) with the true sky brightness temperature field, and the posterior mean of the spectral index $\langle \beta \rangle$ (bottom left) and its fractional difference with respect to the true values (bottom right) with multi-band data between 1.55 to 2.05 GHz.

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